

Floppy Reduction

Rodney Kinney, March, 2023

This method is the result of in-depth discussions about Floppy Reduction on the [Discord server](#).

Credit to Jay, Aneurin, bpaul, ECPmath, Java, Calesch, Robert Yau, and Theodor Nordstrand.

Accompanying videos: [▶ HTR to FR: Using Floppy Reduction to find optimal half-turn solutio...](#)

[▶ Floppy Reduction Update: Recognition and Move Count](#)

Explanation

The [floppy cube](#) is a 1x3x3 version of the Rubik's cube. A 3x3 cube is in "Floppy Reduction" (FR) if it can be solved using only $\langle R2, L2, F2, B2 \rangle$ moves. In FMC, it can be a useful final step after HTR. This document introduces some techniques for going from HTR to FR, for those cases where the finish after HTR is not obvious. The E slice is ignored throughout.

To solve the cube following HTR, we must, of course, solve both the corners and the edges. Parity determines whether the number of U2/D2 turns needs to be even or odd. Our strategy will be to consider the corners and edges separately, find solutions that bring each of them into FR independently, and then "align" the solutions to bring the entire cube into FR.

Cube State Recognition

In FR, a "bad" corner or edge is one with a B sticker on the F face, or an L sticker on the R face. Opposite faces will always have the same number and orientation of bad pieces, so we only need to look at the F and R faces.

Parity

Parity determines whether the number of U2 moves needed to solve the cube is even or odd. The easiest way to determine parity is to count the number of U2 moves needed to solve the corners (up to rotation), then add the parity of the E slice (after corners are solved).

Corner Cases

To recognize corner cases, consider pairs of corners with the same side colors (e.g. UFR and DFR). Looking at any single pair is enough to determine the corner case. We label corner cases by the number of bad corners, and show solutions for even and odd parity.

	0c	4c	6c
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Even parity	-	F2 U2 F2 U2	U2 F2 U2
Odd parity	U2 F2 U2 F2 U2	U2	F2 U2

Similar to DR, we can label each HTR with the number of bad corners plus the number of U2 moves needed to reach Floppy Reduction

		
0c0 0c3	4c1 4c2	6c1 6c2

When we enter HTR, we always have the choice of doing so with either a U or a U' move, so you can cut the number of corner cases in half. It is always possible to force 0c instead of 4c, and for 6c it is always possible to force odd parity. Therefore, it is enough to consider just three corner cases:

		
0c0	0c3	6c1

Edge Cases

Excluding rotations, there are 8 edge cases:

0	4-0	3-1	2-2 slashes	2-2 bars	2-2 slice	1-1	2-0
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Solutions

0c0

For 0c corners, the intuition is to get to 4cp1 corners with 4 bad edges, then bring the bad edges to the U face. The best edge cases are easy to recognize, since all of the bad edges are in the same slice.

-	U2 F2 B2 U2 U2 R2 L2 D2	U2 R2(w) F2 R2(w) U2	F2 U2 R2(w) F2(w) R2 U2

For completeness, here are the solutions for the other edge cases, although in practice, you will typically look for a different HTR.

U2 F2 U2 F2 R2 U2 R2 U2	U2 R2 F2 R2 F2 R2 F2 U2	U2 R2 U2 L2 F2 U2 L2 B2 U2	U2 R2 U2 R2 F2 U2 R2 F2 U2

0c3

For this corner case, the strategy is to start with U2 to get to 4c1, then set up one of these two even-parity triggers:

	+	 U2 F2 U2	 U2 R2 F2 U2
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Interestingly, the best edge cases for 0c3 are also the ones in which the bad edges are in the same slice.

 U2 R2 U2 R2 U2	 U2 R2(w) U2 F2(w) R2 U2	 U2 R2 F2 U2 F2 R2 U2 U2 F2 R2 U2 R2 F2 U2	 U2 R2 F2(w) U2 B2(w) R2 U2
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The remainder of the edge cases work similarly, but require more moves to set up the bad edges

 U2 R2 U2 R2 F2 R2 F2 U2	 U2 R2 U2 F2 R2 F2 R2 F2 U2	 U2 F2 R2 F2 R2 U2 F2 R2 U2	 U2 R2 U2 L2 F2 R2 F2 D2
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6c1

For 6c, the relative alignment of the corners and edges is important. The examples below are for F2 U2 corners. Edges are “not aligned” if they need R2 U2 to be solved.



The intuition is to get to 6c1 with 4 bad edges (2 on each slice), and bring them to the U face. If you start with 4 bad edges, you can set them up immediately:

 F2 U2	 R2 F2 U2	 F2 R2(w) F2(w) R2 U2	 F2 R2(w) B2(w) R2 U2	 R2 F2 R2 F2 R2 U2
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If you don't have 4 bad edges, start with U2 to get to even parity, then set up one of the even-parity triggers from the 0c3 case. You may need to use corner swaps to align the corners and edges.

 U2 R2 U2 R2 F2 U2	 U2 F2 R2 U2 R2 U2	 U2 R2 F2 R2 U2 F2 R2 U2	 U2 F2 R2 F2 R2 F2 U2 F2 U2	 U2 F2 R2 U2 B2 R2 U2	 U2 F2 R2 U2 F2 R2 U2
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Cale Schoon worked out the complete algorithm set in [this spreadsheet](#)

Examples

#1

Scramble: L2 B2 F2 R2 F2 D2 R2 L2 B2 U2 R2 U2 L2 R2 D2 B2 D2 L2 R2 B2
F2 U2 D2 L2 U2

First, we execute U2 to get 0c corners. To see that parity is odd, solve corners with B2 L2 and note that the E slice has odd parity (F2 to solve).

Corner case is 0c3, edge case is 1-1

U2 // 0c3
U2 F2 U2 // 4c1 + 4e
R2 B2 U2 // FR
L2 B2 // Finish

#2

Scramble: L2 U2 F2 L2 R2 B2 L2 D2 F2 D2 U2 B2 U2 F2 D2 R2 U2 F2 R2 F2
L2 D2 R2 L2 B2

Corners are 6c. Parity is even (R2 F2 U2 solves corners, leaving odd parity in the E slice).

```
U2 // 6c1
U2 F2 R2 U2 // 6c1 + 4e
B2 L2 D2 // FR
L2 B2 L2 B2 L2 // Finish
```

The 5-move finish is a clue we might want to keep looking. In fact, the optimal solution uses 4 U2 moves instead of 2, and is one move shorter.

#3

Scramble: R2 D2 U2 L2 D2 B2 L2 D2 R2 U2 F2 L2 B2 U2 F2 B2 L2 D2 F2 B2
L2 R2 U2 B2 D2

Corners are 6c. Parity is odd. Since we have 2 bad edges, we need to make 2 more bad edges using 2 U2 moves and returning to 6cp1.

```
U2 L2 F2 L2 U2 // 6c1 + 4e
B2 R2 D2 // FR
F2 L2 B2 // Finish
```

Pseudo-Floppy Reduction

Accompanying video: <https://www.youtube.com/watch?v=Bakb1FDeneA>

Instead of fully solving the cube from HTR, it can be more efficient to solve only corners and enough edges to leave 2e, 3e, or 2e2e unsolved, then find an insertion for the unsolved edges earlier in the skeleton. This can take many fewer moves than a full solve, and is much faster to find. The strategy is simply to ignore parity and ignore either the M or S slice. Some examples:

No Setup

These HTR scrambles land already in Pseudo-FR, or require just a U2 move.

```
L2 R2 F2 B2 R2 U2 R2 D2 U2 B2 U2 L2 F2 B2 R2 U2 L2 F2 L2 D2 U2 R2 B2 D2 R2
```

```
// Pseudo-FR (ignoring M slice)
L2 F2 B2 R2 B2 // 3e
```

```
B2 R2 L2 D2 B2 F2 U2 F2 R2 D2 L2 D2 B2 F2 L2 R2 D2 L2 F2 B2 L2 U2 R2 F2 B2
```

```
U2 // Pseudo-FR (ignoring M slice)
```

F2 B2 R2 L2 // 2e

R2 F2 L2 U2 B2 U2 D2 R2 D2 U2 F2 D2 R2 L2 F2 R2 F2 L2 R2 F2 D2 L2 F2 B2 R2

U2 // Pseudo-FR (ignoring S slice)

B2 R2 B2 R2 // 3e

Short Setup

These scrambles require a side move plus a U2 to bring the corners into FR, where the side move can be chosen to also bring one of the slices into FR.

D2 U2 B2 R2 L2 F2 D2 U2 L2 U2 L2 F2 R2 L2 F2 L2 U2 D2 F2 L2 F2 R2 F2 D2 F2

R2 D2 // Pseudo-FR (ignoring S slice)

R2 B2 R2 // 2e

L2 F2 B2 U2 B2 R2 F2 R2 U2 L2 F2 L2 D2 R2 B2 D2 B2 D2 F2 L2 D2 L2 U2 L2 R2

F2 U2 // Pseudo-FR (ignoring M slice)

F2 L2 F2 R2 F2 // 3e

Corner Swap setup

If an edge is in the wrong layer to be brought into FR along with the corners, then a corner swap can be used to move it between the U and D layers.

U2 F2 U2 R2 F2 R2 L2 B2 F2 R2 F2 L2 B2 R2 F2 R2 B2 R2 B2 F2 L2 B2 F2 D2 R2

F2 L2 F2 U2 // Pseudo-FR (ignoring M slice)

R2 // 3e

B2 R2 L2 F2 U2 F2 L2 B2 F2 L2 F2 B2 D2 L2 R2 B2 L2 D2 B2 D2 F2 L2 R2 F2 U2

R2 B2 R2 B2 D2 // Pseudo-FR (ignoring M slice)

L2 B2 // 3e

F2 R2 B2 R2 B2 D2 // Pseudo-FR (ignoring M slice)

L2 // 3e

Triple Slice

All of the above solutions have the unsolved edges in a single slice. It is also possible to use Pseudo-FR to get to 2e2e with two bad edges in each slice.

U2 B2 D2 F2 L2 U2 B2 D2 U2 F2 R2 U2 D2 R2 U2 B2 U2 D2 F2 U2 R2 U2 B2 F2 R2

// Pseudo-FR (FR -2e2e)

R2 L2 F2 R2 L2 // 2e2e

L2 R2 D2 L2 U2 F2 U2 D2 B2 U2 D2 L2 B2 L2 D2 U2 F2 R2 U2 F2 D2 R2 U2 D2 B2

U2 L2 F2 U2 // Pseudo-FR (ignoring M slice)

R2 // 2e2e